CSE421 Assignment 03 [MSMA | Summer 2025]

Answer all the questions in this form. You can submit only once even if you submit by mistake. Make sure that your answers are put correctly by refreshing the page.

Q1. Which of the followings are true?

1. TCP is not faster than UDP even though we can see websites get loaded within seconds
2. DNS uses both TCP and UDP as transport layer protocol
3. Both TCP and UDP initiates the connection through 3 way handshaking
4. Reassembling the segments are done by both TCP and UDP
5. Reordering the segments are done by both TCP and UDP
6. None of the above

Answer: 1,2,4

Answer Q2 to Q3 based on the following scenario.

I have opened two tabs in my browser and requested to visit the same webpage. Then the server responded with the webpage.

Q2. Source port address of the request packet can be from (Select all possible answers evaluating different scenarios)

1. Dynamic
2. Registered
3. Well known
4. 1024 to 49151

5. 49152 to 65535
6. 0 to 1023

Answer: 1,5

Q3. Destination port address of the request packet can be from

1. Dynamic
2. Registered
3. Well known
4. 1024 to 49151
5. 49152 to 65535
6. 0 to 1023

Answer: 2,3,4,6

Answer Q4 to Q5 based on the following scenario.

There are 2 computers that are requesting to a server's process. Among these computers, PC1 opens **4 connection**, PC2 opens **9 connections** with the server.

Q4. If the protocol used is UDP, how many sockets will be created at the server side to send the message to application layer?

Answer: 1

Q5. If the protocol used is TCP, how many sockets will be created at the server side to send the message to application layer?

Answer: 13

Answer Q6 to Q9 based on the following scenario.

A segment sent by the client has size of **670 Bytes** including **56 Bytes** of header. The sequence number is **8122** and acknowledgement number is **3002**. The urgent flag is set and there are **120 Bytes** of non-urgent data.

Q6. What will be placed on HLEN field of the header? (in Binary)

Answer: 1110

Q7. What will be the value of the urgent pointer? (in Decimal)

Answer: 493

urgent data = 670 - 56 - 120 = 494

last urgent byte = seq + urgent data - 1

last urgent byte - seq = urgent data - 1

urgent pointer = 494 -1 = 493

Q8. What will be the value of the first non urgent byte number? (in Decimal)

Answer: 8616

urg flag = 1 so, there is urgent data.

last urgent byte = 8122+493

first non urgent = 8122+493+1 = 8616

Q9. Which of the followings are True?

1. The client expect to receive from byte Number 8122
2. The client has received 8122 Bytes of Data
3. The client has received 3002 Bytes of Data
4. The client expect to receive from byte Number 3002
5. None of the above

Answer: 4

Q10. Suppose the client has received only one in order segment. What will the client do?

1. Wait till it's timer is finished before sending ack
2. Wait till the server's timer is finished before sending ack
3. Wait for another segments to arrive since it has received only one segment and then send ack
4. immediately send an ack

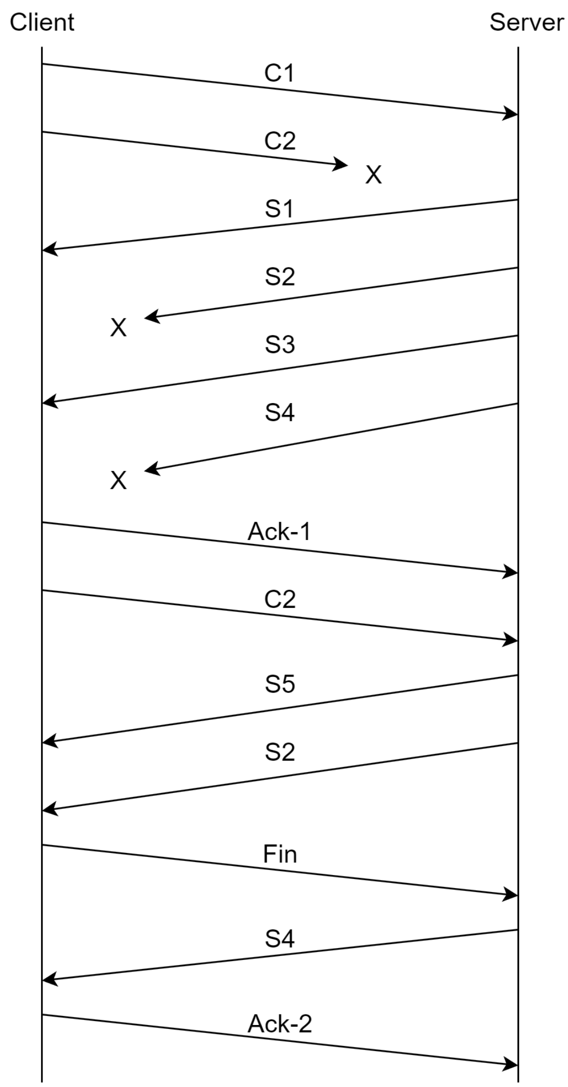
Answer: 1,3

After the three-way handshaking, the client has sent the **C1** segment. The ISN of the client is **5000** and the ISN of the server is **7000**. The RWND of the client is **16000 bytes** and the RWND of the server is **20000 bytes** at the initial stage. The size of the segments **C1, C2, S1, S2, S3, S4,** and **S5** are **421, 260, 320, 422, 110, 321,** and **230 bytes** respectively.

Sanity Checking: You can't use this value to solve the math. However, you can check whether you are in the right track or not by matching your answer with this.

Seq of S5 = 8174

Ack of S5 = 5682



Q11. Find out which sliding window protocol has been used here.

1. Selective Repeat
2. Go-Back-N

Answer: 1

only the missing segments were resent

Q12. Determine the sequence number of the segment labeled as 'Ack-1'.

Answer: 5682

Seq of C1 = 5000+1(for syn packet) = 5001

Ack-1 is after C2.

5001+421+260=5682

Q13. Determine the acknowledgement number of the segment labeled as 'Ack-1'.

Answer: 7321

Seq number of S1 = 7000+1(for syn-ack packet)=7001

It will ask to send from S2.

seq of S2 = 7001+320=7321

Q14. Determine the sequence number of the second/sent 'S4' segment.

Answer: 7853

S4 is after S1, S2 and S3,

so, seq of S4 = 7001+320+422+110=7853

Q15. Determine the acknowledgment number of the second/sent 'S4' segment.

Answer: 5683

received upto c1, c2, fin segment fully before sending the second S4,

ack of second S4 = 5001+421+260+1=5683

Q16. Determine the acknowledgment number of the segment labeled as 'ACK-2'.

Answer: 8404

received and stored s1, s3, s5, s2, s4 segments before sending ack-2. so all segments till s5 are received.

ack will be after s5,

7001+320+422+110+321+230=8404

Q17. Determine the value of Sn for the server's sending window after sending the second 'S2' segment.

Answer: 8404

Server has already sent upto S5. so, new data will be after S5.

7001+320+422+110+321+230=8404

Q18. Determine the value of Sf for the server's sending window after sending the second 'S2' segment.

Answer: 7321

Sf = first byte which is yet to be acknowledged or, last acknowledgement got from the receiver

so, acknowledgment of received C2 is Sf

ack of received C2 = ack of Ack-1 = 7321

Q19. Calculate the window size of the server after receiving the 'Fin' segment if it could process 'C1'.

Answer: 19740

Received Syn, C1, C2 and Fin after declaring rwnd as 20000. but fin/syn segment is not stored in the buffer.

20000-421-260+421(processed the C1 segment)=19740

Q20. Calculate the window size of the 'Fin' segment.

Answer: 14918

Fin segment is sent by the client, so it will hold the client's receiving window in it's window field.

Client received Syn, S1, S2, S3, S5 after declaring rwnd as 16000. Stored all segments in the buffer except the Syn segment.

16000-320-422-110-230=14918